

Solving Word Problems Involving Volume of Pyramids

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The **volume** of any pyramid is one-third the volume of a prism with the same base and height.

Solving Word Problems

Read the problem carefully, identify the given measurements, compute the base area first, then apply the volume formula.

Cubic Units

Always write the final answer in **cubic units**.

Volume = cubic units

Examples

Example 1

A square pyramid has a base with side length 12 cm and a height of 15 cm. Find its volume.

Example 2

A rectangular pyramid has a base measuring 10 cm by 8 cm and a height of 18 cm. Find its volume.

Example 3

A monument is shaped like a square pyramid. The base side is 25 m, and the height is 18 m. Find the volume.

Practice Exercises

Practice 1

A square pyramid has a base with side length 12 cm and a height of 15 cm. Find its volume.

Practice 2

A rectangular pyramid has a base measuring 10 cm by 8 cm and a height of 18 cm. Find its volume.

Practice 3

A square pyramid has a base side of 14 m and a height of 9 m. Determine its volume.

Practice 4

A rectangular pyramid has a base measuring 16 m by 9 m and a height of 12 m. Find its volume.

Practice 5

A monument is shaped like a square pyramid. The base side is 25 m, and the height is 18 m. Find the volume.

Practice 6

A flower bed is shaped like a rectangular pyramid with a base measuring 7 m by 4 m and a height of 6 m. How much soil can it hold?

Practice 7

A solid toy is shaped like a square pyramid with a base side of 9 cm and a height of 8 cm. Find its volume.

Activities

Activity 1

A square pyramid has a base with side length 15 cm and a height of 12 cm. Find its volume.

Activity 2

A rectangular pyramid has a base measuring 12 cm by 9 cm and a height of 15 cm. Find its volume.

Activity 3

A square pyramid has a base side of 16 m and a height of 15 m. Determine its volume.

Activity 4

A rectangular pyramid has a base measuring 18 m by 10 m and a height of 14 m. Find its volume.

Activity 5

A pyramid has a rectangular base measuring 24 cm by 18 cm and a height of 9 cm. How many cubic centimeters does it contain?

Activity 6

A decorative pyramid has a square base with side length 20 m and a height of 24 m. Find its volume.

Activity 7

A storage container is shaped like a rectangular pyramid. Its base measures 8 m by 6 m, and its height is 10 m. What is its volume?

Thank you!